

Ranking Transfers, Thresholds Don’t: Prior-Corrected Test-Time Adaptation for Audio Deepfake Detection

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Abstract—Audio deepfake detectors generalise poorly across corpora despite near-perfect in-domain accuracy. We show the failure is one of *operating point*: a fine-tuned self-supervised detector’s *ranking* (ROC-AUC) largely transfers cross-corpus while its decision *threshold* does not, so unlabeled target audio can repair the operating point at test time. We then show that confident-quantile test-time adaptation (TTA) can be sound only while its tail budget satisfies $q \leq \min(\pi, 1-\pi)$: a counting bound caps pseudo-label purity at $\min(1, \pi_{\text{real}}/q)$ for any detector. That holds where TTA is measured and essentially nowhere else — ASVspoof2021-DF is 97.2% spoof — so at the published $q=0.3$ no model can exceed 9.3% purity there. Resampling In-the-Wild’s prevalence, holding audio and weights fixed, reproduces the failure on demand: accuracy tracks the bound across a 3×3 grid (Spearman $+0.966$). Empirically, on the official DF eval (400,435 trials) that configuration destroys the operating point of three released state-of-the-art checkpoints, driving accuracy from 98.8–99.1% to 62–65% across nine paired cells; three of four published TTA baselines degrade all three as well — ETA reaching 31 and 37% EER against a source 3.8 and 4.6 — each through its own version of the assumption. Correcting the budget with a label-free prior estimate ($\hat{\pi}=0.9715$ vs. a true 0.9708) restores all of them and beats the symmetric budget on 9/9 cells ($p=0.004$); both EER and AUC move, so no re-thresholding rule accounts for it. Its boundary we measure rather than assume: the prior is not identifiable from a model’s own outputs, so the estimate degrades exactly as the operating point does ($r=0.997$ over ten cells), and a wrong prior is worse than none. Since q is ours to choose and π is not, shrinking q satisfies the condition without estimating anything.

Index Terms—audio deepfake detection, anti-spoofing, test-time adaptation, label shift, cross-corpus generalisation, EER.

I. INTRODUCTION

Self-supervised (SSL) front-ends such as wav2vec 2.0 and XLS-R, paired with lightweight back-ends, reach very low in-domain error on standard anti-spoofing benchmarks [1]–[3]. Deployed on audio from a different source—new microphones, codecs, languages, or generators—accuracy drops steeply, a well-documented generalisation gap [5], [6], [17].

We start from a diagnostic measurement: a fine-tuned XLS-R detector evaluated on an unseen corpus keeps a high ROC-AUC while its accuracy at a fixed threshold collapses. It still *ranks* clips correctly; its operating point is mis-calibrated. Because ranking transfers, the most-confident predictions on

unlabeled target audio are trustworthy enough to adapt on at test time.

That observation is the starting point of a family of methods, not a method. The step every member of the family takes—label the extreme quantiles of the target scores and self-train on them—carries an assumption that is invisible on the benchmarks the field uses and fatal on the pools it deploys to: that the target pool is roughly class-balanced. Take the top and bottom 30% of a 97%-spoof pool and the “confidently bona fide” bucket is mechanically made of low-scoring *spoof*. Self-training then teaches the detector that a great deal of spoof is genuine. This is not hypothetical: it is what happens to all three released state-of-the-art checkpoints we test on the official ASVspoof2021-DF protocol (Table I), and to three of four modern TTA baselines.

Contributions. (i) The ranking–calibration asymmetry as the mechanism of cross-corpus failure, over 369 third-party cells and tested directly: a proxy $\mathcal{A}$ -distance *rises* after adaptation, so the effect is operating-point repair, not learned invariance. (ii) An exact validity condition for the confident-quantile rule, $q \leq \min(\pi, 1-\pi)$, from a counting bound on pseudo-label purity; it predicts every result here, including why three of four modern TTA baselines destroy three released SOTA checkpoints at the standard benchmark’s own prevalence, and it is confirmed by intervention. (iii) A label-free correction that removes the damage on every DF checkpoint, together with an identifiability argument bounding when it can be trusted — and the demonstration, via an oracle prior, that what fails is the estimate and not the correction. (iv) Two label-free monitors that separate damaging adaptations from safe ones, one of which recovers 13.9 EER points from a Tent collapse out-of-sample.

Related work. SSL front-ends with AASIST or attentive back-ends dominate recent benchmarks [1]–[4]. Train-time domain generalisation aligns source domains [5] or runs gradient reversal over source and target jointly [13]; both act before deployment. In TTA, entropy minimisation (Tent) [7] is the reference recipe, and its instability prompted reliable-sample filtering (ETA/EATA) [8], sharpness-aware entropy with recovery (SAR) [9], and information maximisation with self-labelling (SHOT) [10] — we compare against all four. Batch-statistics recalibration [14] is a no-op for LayerNorm.

The prior estimator is black-box shift estimation [11]; the proxy $\mathcal{A}$ -distance is from [15].

II. METHOD

Let x be a waveform with label $y \in \{0, 1\}$ (fake= 1) and let f_θ produce $s_\theta(x) = P_\theta(y=1 | x)$. We are given a source model, its labelled training corpus $\mathcal{S}$, and an *unlabelled* target pool $\mathcal{T}$. No target label is read at any point.

What breaks under corpus shift. Ranking quality AUC = $\Pr[s_\theta(x^+) > s_\theta(x^-)]$ is threshold-free; accuracy at a fixed τ is not. Empirically AUC degrades far less than accuracy at $\tau=0.5$ under source→target shift: across 369 third-party evaluation cells the accuracy a model’s ranking supports but its shipped threshold does not deliver averages 7.29 points. The deployable object is therefore the score *order*, and the broken object is the operating point. Everything below follows from taking that literally.

(a) The target prior, label-free. An operating point is a statement about a prior, so we estimate the prior rather than assume it. Black-box shift estimation [11] gives $\hat{p} = M^{-\top} q$ with $M[y, \hat{y}] = P_{\mathcal{S}}(\hat{y} | y)$ the source confusion matrix and q the target prediction histogram. Estimators split into those reading the *thresholded prediction rate* (BBSE, EM [12]) and those reading the score *histogram* (a two-component mixture); the first inherits whatever calibration error the target induces, and Secs. IV and V show neither is reliable where it is needed.

(b) The tail budget, and when it is valid. The standard rule takes the bottom q and top q of the target scores as pseudo-real and pseudo-fake. Its validity has a characterisation that needs no model at all. A bucket of $q_c N$ clips can contain at most the pool’s $\pi_c N$ class- c clips, so *for any detector whatsoever*

$$\text{purity}_c \leq \min(1, \pi_c / q_c). \quad (1)$$

This is counting, not an approximation; what is empirical is its *tightness*, which is where the ranking precondition enters — across all seven (checkpoint, corpus) cells here the bound is never violated and is attained to within 0.025, the slack tracking AUC (0.000 at AUC .985, 0.025 at .917). Two consequences follow, and they are the whole story.

The *symmetric* rule sets $q_c=q$ and is therefore sound only while

$$q \leq \min(\pi, 1 - \pi). \quad (2)$$

In-the-Wild satisfies this at $q=0.3$ (min=.372) and the recipe repairs checkpoints there; DF does not (min=.028), where no detector can exceed 9.3% pseudo-real purity — we measure 8.6% — and the recipe destroys them. The damage is one-sided: DF’s pseudo-fake tail is 100% pure at every q .

The *prior-split* rule, $q_c=2q\hat{\pi}_c$, restores Eq. (1) for *any* $q \leq \frac{1}{2}$ given a correct prior, and is identical to the symmetric rule at $\hat{\pi}=0.5$. With an estimate it needs

$$\hat{\pi}_c / \pi_c \leq 1/(2q) \quad \text{for both classes}, \quad (3)$$

i.e. at $q=0.3$ the estimate may overstate either class by at most $1.67\times$. This is why it works on DF ($\hat{\pi}_{\text{real}}/\pi_{\text{real}}=1.03$) and fails on In-the-Wild s42, where underestimating π_{real} inflates

the *fake* budget to 0.587 against $\pi_{\text{fake}}=0.372$ ($2.63\times$, bound 0.633) and simply moves the impurity to the other tail. Across the seven adapted arms we ran, the worst-tail bound is 1.000 for the five that came out intact (82–99% accuracy) and 0.093 and 0.633 for the two that collapsed (62.96, 47.99). Since q is ours to choose and π is not (Sec. V), a third option is to shrink q until Eq. (2) holds for the most skewed prior anticipated — at a supervision cost the prior-split budget avoids ($q=0.02$ labels 800 of 20,000 clips against 12,000). A fixed confidence threshold is a fourth, prior-free but calibration-dependent. We report all four.

(c) Objective. With pseudo-labels $\tilde{y}$ from (b) on the confident set $\mathcal{C}$ and a stochastic channel perturbation $a(\cdot)$ (per-clip gain $g \sim \mathcal{U}(0.7, 1.3)$ plus Gaussian noise $\sigma=0.005$),

$$\min_{\{\gamma, \beta\}, \phi} \underbrace{\sum_{x \in \mathcal{C}} \text{CE}(f_\theta(x), \tilde{y})}_{\mathcal{L}_{\text{ST}}} + \lambda \underbrace{\|f_\theta(a(x)) - \text{sg}[f_\theta(x)]\|^2}_{\mathcal{L}_{\text{cons}}}, \quad (4)$$

optimising only the LayerNorm affine parameters $\{\gamma, \beta\}$ of the fine-tuned blocks and the head ϕ (16,706 parameters), $\lambda=0.3$. Restricting the trainable set and using only prior-corrected confident tails is what prevents the degenerate collapse entropy minimisation is prone to. The consistency term is inherited from the published recipe; an ablation finds it contributes nothing here (3.98 vs 4.02 EER), so the budget is doing the work.

(d) Two label-free guards. Adaptation can damage the model it is given, so a deployer needs to decide — without labels — whether to keep the adapted weights. Since the source ranking is the asset being exploited, we compare the adapted scores’ order against the frozen source model’s, $\rho = \text{Spearman}(s_\theta(\mathcal{T}), s_{\theta_0}(\mathcal{T}))$, and separately track the drift of the predicted positive rate from $\hat{\pi}$. We use these *post hoc*, as a keep/abort decision on a completed adaptation; a per-step stopping rule is the natural extension and we do not claim it here.

III. EXPERIMENTAL SETUP

Protocol A. The official ASVspoof2021-DF eval partition against the official CM keys; 400,435 of 611,829 trials are held locally, the missing part uniform across codec, label and phase (verified 75–25 in every stratum), true spoof rate 97.22%. Published pooled EERs here run from 1.95 (Wav2DF-TSL) and 2.87 (SSL+SLS) through 8.54 (Wav2Vec2-AASIST) to 15.64 (challenge top-1); our own source model reaches only 26.33, so we run the method on *released* checkpoints instead: DeepFense XLS-R 300M+AASIST trained on ASVspoof2019-LA [22], three independent training seeds. Adaptation uses an unlabelled 20,000-clip subsample; we report the full official pooled EER, an inductive number over trials *not* in that subsample, and two further seeds on a fixed 50,000 subset, giving nine paired cells.

Protocol B. Leave-one-corpus-out over ASVspoof2019-LA [18], D2 (LibriSpeech vs. eight open TTS systems), In-the-Wild [17] and the Arabic set [20]; each source pool merges

TABLE I

OFFICIAL ASVSPOOF2021-DF EVAL, 400,435 TRIALS, 97.22% SPOOF, RELEASED XLS-R+AASIST CHECKPOINT SEED 2 [22]; EVERY ARM WAS ALSO RUN ON SEEDS 42 AND 240 (TEXT). ALL ADAPTATION IS LABEL-FREE. DEFICIT $\delta = (100 - \text{EER}) - \text{acc}@0.5$ IS THE ACCURACY THE RANKING SUPPORTS BUT THE THRESHOLD DOES NOT DELIVER. ρ AND THE PREDICTION-RATE GAP ARE THE LABEL-FREE MONITORS OF SEC. II(D) — COMPUTABLE AT DEPLOYMENT, UNLIKE EVERY OTHER COLUMN.

	EER %	AUC	acc@0.5	deficit	ρ	gap
Source (as released)	4.51	.9923	98.87	-3.38	1.000	.000
+ median-threshold rule	4.51 [†]	.9923 [†]	59.06	—	—	—
+ ETA [8]	7.44 [‡]	.9288	98.93	-6.37	0.312	.007
+ Tent [7]	4.06 [‡]	.9765	98.52	-2.58	0.438	.013
+ SHOT [10]	6.38	.9725	77.64	+15.99	0.767	.225
+ TTA, fixed confidence 0.95	4.30	.9924	99.02	-3.32	0.827	.005
+ TTA, symmetric $q=0.3$	5.84	.9846	62.96	+31.20	0.816	.372
+ TTA, prior-corrected	4.02	.9931	98.95	-2.97	0.867	.007
(disjoint/inductive half)	4.03	.9931	98.95	—	—	—
(no consistency term)	3.98	.9936	98.90	-2.88	0.884	.003
+ SAR [9]	4.47 [§]	.9923	98.88	-3.35	0.970	.000

[†] A threshold cannot change EER or AUC; both come from the score ROC. The median rule is a control for the *accuracy* claim and cannot produce any EER result in this paper. [‡] EER over a pool that is more than 93% a single tied score (Tent saturates 98.07% of scores to exactly 1.0, ETA 97.71%); read AUC. [§] SAR neither damages nor adapts: 308 recovery resets in 2,500 steps, AUC unchanged to four decimals.

the other three with a 38-language MLAAD subset [19], held-out target audio never appears in any source pool, and six MLAAD languages are held out of every pool. Every method runs on the same twelve folds (four targets, three seeds).

Protocol C. All 31,779 In-the-Wild clips at their native 37.2% fake rate, skewed the *opposite* way to DF.

Comparison points. Source-only; the median-threshold rule; Tent, SHOT, ETA/EATA, SAR; AdaBN [14]; DANN [13] (which sees the held-out target’s unlabeled audio every step, more privileged access than ours); an ASDG-style baseline [5]; and a target-supervised oracle. Every TTA arm adapts the same 16,706 parameters with the same lr, batch and pass count; none was tuned separately, and none could be, since per-target tuning needs the labels the setting assumes absent. Two published constants are degenerate at $C=2$ and were re-derived: EATA’s redundancy filter ($|\cos| < 0.05$ is unreachable until the running mean hits a simplex vertex, after which only minority-predicted samples pass) and SAR’s recovery trigger ($e_0=0.2$ sits inside the $C=2$ reliable band and freezes the model; we keep its relative position, $e_0=(0.2/\ln 1000)\ln C$).

Implementation. 16 kHz audio, top-4-layer XLS-R fine-tuning for our own source models (Adam, $\text{lr} 2 \times 10^{-4}$, batch 32, 8 epochs). Adaptation: $\text{lr} 1 \times 10^{-4}$, $E=4$ passes, batch 32 (third-party crops matched to each checkpoint’s own training length). Code and manifests: [URL withheld].

IV. RESULTS

A. The class prior is the hidden variable

Table I is the main result, and Eqs. (1)–(2) anticipate it. Applied unmodified to a released checkpoint already better than the published Wav2Vec2-AASIST baseline here, the symmetric-quantile configuration *damages* it: accuracy $98.87 \rightarrow 62.96\%$, EER $4.51 \rightarrow 5.84\%$, manufacturing a 31-point deficit where there was none.

The published baselines are no better placed, and each fails through its own version of the assumption. SHOT minimises $\mathbb{E}[H(p)] - H(\mathbb{E}[p])$, whose diversity term is maximised when the mean prediction is *uniform* — at $C=2$ a hard prior of $\frac{1}{2}$; the regulariser introduced to *cure* Tent-style collapse encodes the assumption more literally than Tent does. ETA fails more instructively: its reliability filter, meant to discard the high-entropy samples that destabilise entropy minimisation, admitted 99.8% of them, so ETA reduced to Tent and landed worse. An entropy-keyed filter separates confident from uncertain, not *confident-and-right* from *confident-and-wrong*, and mis-calibration manufactures exactly the latter. SAR is the exception that clarifies the rule: the only prior method that does not degrade the checkpoint, and it manages that by declining to move it (308 recovery resets in 2,500 steps, AUC unchanged to four decimals). Across all three checkpoints ETA reaches 31.0 and 37.5% EER (source 3.8, 4.6), Tent 17.6 and 11.2, SHOT costs 17–26 accuracy points on every one, and SAR is inert on every one. So the claim is not that every published method fails here, but that *three of five prior configurations degrade a state-of-the-art detector at the standard benchmark’s own prevalence, the fourth avoids damage only by declining to adapt, and none improves it*.

The prior-split budget reverses this: BBSE returns $\hat{\pi}=0.9715$ against a true 0.9708, comfortably inside Eq. (3), and EER improves to 4.02%. The change between the two arms is one scalar. Over three independently trained checkpoints (EER % / accuracy %):

	seed 2	seed 42	seed 240
source	4.51 / 98.87	4.56 / 98.78	3.82 / 99.11
+ symmetric q	5.84 / 62.96	4.08 / 61.80	4.66 / 65.24
+ prior-corrected	4.02 / 98.95	3.29 / 98.98	3.80 / 98.74

Over the nine paired cells (seed-to-seed EER std ≤ 0.14), prior-correction beats the symmetric budget on 9/9 (+1.23 EER, Wilcoxon $p=0.004$) and source-only on 8/9 (+0.48, $p=0.027$), while the symmetric budget is worse than source-only on 6/9. Accuracy separates them with no overlap at all: 98.8–99.1% (source), 61.6–65.2 (symmetric), 98.7–99.0 (prior-corrected). What does *not* replicate is the symmetric budget’s effect on *ranking* (EER worse on two checkpoints, better on one), as Eq. (1) implies: a mis-specified budget corrupts the pseudo-labels and hence the operating point, while the ranking is inherited.

B. The failure, and where the recipe already works

At In-the-Wild’s native prior (0.372) Eq. (2) *holds*, and the same recipe repairs rather than destroys. DeepFense s42 — the checkpoint that scores 4.56% on DF — arrives here ranking at AUC 0.917 while correct on only 39.1% of clips, fewer than a majority-class predictor. Self-training repairs exactly that: over four released checkpoints it drives the deficit from 16.7, 22.6 and 44.4 to 0.1, 2.9 and 5.0 (and costs a little on the fourth, which arrives well-calibrated at 2.1 and has nothing to repair). On s42 it also raises AUC .917 $\rightarrow$.947 — which no threshold

can do — with the disjoint half slightly better (12.27/.949), so the ranking gain is not memorisation.

C. Testing the bound by intervention

Eqs. (1)–(2) are claims about the budget, so we move the budget. Sweeping q on DF s2 with everything else fixed gives accuracy 98.96, 95.68, 88.15, 62.96 at $q=0.02, 0.05, 0.10, 0.30$ against bounds 1.000, 0.556, 0.278, 0.093 — monotone in the bound (Spearman +1.000). At $q=0.02$, inside Eq. (2) for this pool, the *unmodified symmetric rule* is safe and improves on source (4.51 $\rightarrow$ 4.19) — no prior estimate anywhere. It remains weaker than the prior-split budget (4.02) for the reason Eq. (1) implies: shrinking q fixes a minority-tail problem by discarding the majority tail too (800 of 20,000 clips labelled, against 12,000).

Inducing the failure where the recipe works. DF is one benchmark that happens to be 97.2% spoof, so we intervene on prevalence directly: resample In-the-Wild — where $q=0.3$ satisfies Eq. (2) and repairs checkpoints — to a target $P(\text{fake})$, holding audio and weights fixed. Accuracy, bound in parentheses:

$P(\text{fake})$	$q=0.02$	$q=0.10$	$q=0.30$
0.90	97.8 (1.00)	95.9 (1.00)	73.3 (0.33)
0.95	98.6 (1.00)	92.6 (0.50)	68.5 (0.17)
0.97	98.6 (1.00)	89.4 (0.30)	65.4 (0.10)

Accuracy tracks the bound (Spearman +0.966): $q=0.3$ collapses monotonically, $q=0.10$ degrades from 0.95 exactly where its bound does, and $q=0.02$ is unharmed. The failure belongs to prevalence, not to DF.

Is the correction wrong, or the estimate? On the hardest cell (In-the-Wild s42, $\delta=44.4$) we substitute the *true* prior:

prior used	none (sym.)	BBSE	mixture	true
EER / accuracy	12.58 / 82.4	14.51 / 48.0	14.95 / 55.0	11.59 / 91.3

With the true prior the prior-split budget is the best method in this paper on its hardest cell; with either estimate it is worse than using no prior. The mechanism is sound and the estimation is not — exactly what Sec. V proves must happen.

Not a re-thresholding result. Both EER and AUC move, and neither takes a threshold as input: the same prior estimate used as a threshold leaves EER at 4.51 by construction, and as a budget gives 4.02.

D. Cross-corpus, balanced pools: where the advantage disappears

Here Eq. (2) holds and the prior correction is inert by construction, so this arm measures the rest of the method. On twelve identical folds (four leave-one-corpus-out targets, three seeds) we beat source-only (18.39 $\rightarrow$ 17.44 pooled EER, 10/12, $p=0.027$), Tent (32.25, $p=0.012$), ETA (29.54, $p=0.003$), EATA (31.34, $p=0.016$) and SAR (18.39, $p=0.027$) — but **not SHOT** (17.76 vs our 17.44, 7/12, $p=0.47$), which is better than us on ASVspoof2019. That is the point rather than an embarrassment: SHOT’s diversity term is a hard uniform prior, free when Eq. (2) holds and destructive when it

does not (6.38% EER and 77.6% accuracy on DF against our 4.02 and 98.95). The separation between the two methods lives entirely in the regime the field does not test. Tent, ETA and EATA are unstable even here, and SAR’s pooled EER equals source-only to two decimals — inert on balanced pools exactly as it was on DF.

E. Label-free guards

Whether to keep an adapted model is itself decidable without labels. Over the eight adapted DF arms, rank agreement ρ and prediction-rate drift are complementary — ρ catches ranking collapse (ETA 0.31, Tent 0.44), the drift catches operating-point collapse (0.37, 0.39, 0.23) and is blind to the former — and requiring both separates the five harmful arms from the three harmless ones exactly, for any $\tau \in [0.70, 0.80]$. The plateau matters more than the exactness: a ρ -only rule at $\tau=0.85$, fitted on seven arms, looked equally exact and was falsified by the eighth, rejecting the study’s best result. Two thresholds on eight points is a diagnostic, not a rule, so we validate the ranking guard as a *stopping* rule out of sample: twelve folds ($E=32$), threshold chosen leave-one-target-out. On Tent it recovers 13.9 EER points (32.25 $\rightarrow$ 18.33, oracle 17.24) — back to source level from a catastrophic collapse — while on our own method it never fires and simply runs the budget out (17.44 $\rightarrow$ 16.58, oracle 16.12). It prevents collapse and otherwise does not interfere, rather than selecting a budget; churn and prior-drift are weaker still.

F. Breadth, and the mechanism

Over nine released checkpoints, five architectures and ten corpora (369 cells), the consistent effect is *re-calibration*: δ falls 7.29 $\rightarrow$ 1.96 (248/369, $p=2.2 \times 10^{-17}$), and where it lands barely depends on where it started (corr=+0.17). The mechanism is not learned invariance: a proxy $\mathcal{A}$ -distance and a linear domain-origin probe both *rise* after adaptation (Arabic 1.545 $\rightarrow$ 1.606), consistent with DANN underperforming while optimising that quantity. Beyond any threshold the gradient step adds separability where the ranking is strong: on ASVspoof2019 within-attack-family AUC rises on all six families (30/30, $p=10^{-9}$); on the AUC-0.71 target it is null.

V. LIMITATIONS

The prior is not identifiable, which bounds the correction. A label-free method sees only $\Pr[\hat{y}=1] = \pi \text{TPR}_{\mathcal{T}} + (1-\pi)\text{FPR}_{\mathcal{T}}$: one equation, three unknowns. BBSE closes it by assuming source error rates carry over — what covariate shift breaks — so an extreme prior and an extreme calibration failure look identical (DF/s2 predicts 97.4% fake at a true 0.972, ITW/s42 98.1% at a true 0.372). The estimate therefore degrades exactly as the operating point does — over ten cells and two corpora, $|\hat{\pi} - \pi|$ against the source deficit gives Pearson $r=0.998$ — and a badly wrong prior is worse than none (ITW/s42 12.58 $\rightarrow$ 14.51 EER). Estimating the prior and detecting that the estimate is wrong are one problem; two label-free detectors for it are falsified in our release. The gain also varies a great deal (1.27 to 0.02 EER), and a low-AUC start we cannot rescue.

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